

Joint Depth-Normal Optimization

Novel contribution: First method to jointly optimize metric depth and surface normals through an iterative process.

ConvGRU iterative refinement:

1. Decoder produces initial coarse estimates $D_0, \mathbf{n}_0$
2. At each iteration t :
 - ▶ Compute update Δ_t from hidden state
 - ▶ $D_{t+1} = D_t + \Delta_{D,t}$
 - ▶ $\mathbf{n}_{t+1} = \mathbf{n}_t + \Delta_{\mathbf{n},t}$
3. After T steps: final refined output

Why jointly? Depth and normals are geometrically coupled. Sharing a hidden state lets each task inform the other.

